

ALUMINIUM STORE

TECHNICAL & PRODUCT SPECIFICATION

Version 1.0 · September 2026

A Specialised Operating Division of Besbpo Group

This document adapts the Group's e-commerce blueprint specifically to a configurator-led, single-vendor aluminium platform — with wireframes, architecture diagrams, and the compliance architecture required for the South African market (ECTA, POPIA, PAIA, CPA, SARS).

1. Executive Summary

Aluminium Store's platform is a single-vendor, configuration-heavy e-commerce and B2B system — closer in shape to Bellwether SWE Plumbers and Roofsteel than to a multi-seller marketplace. This specification documents what has been scaffolded: 59 data models, 30 backend modules, 38 frontend routes, and a pricing engine cross-verified against the Pricing Framework workbook to the cent.

What makes this platform different from a generic e-commerce build: a live configurator is central to nearly every SKU, not optional; large curtain-walling and shopfront orders route through a vetted CMI Partner Network with a human-in-the-loop checkpoint; fragile-glazed logistics is a distinct handling class; and every compliance requirement — NRCS VC 9003 safety glass, SARS Tax Invoices, POPIA data-subject rights, PAIA disclosure — is modelled as a first-class feature, not an afterthought.

What's real vs. scaffolded

- Real, working logic: the configurator's live pricing, the six-provider authentication flow with account-linking, CMI partner routing with human confirmation, coupon/discount math, SARS invoice numbering, ECTA non-returnable enforcement, the Business Desk's multi-seat Company model.
- Honestly scaffolded, not faked: the AI support agent's conversation and escalation flow is real; the generative reply itself is marked for Phase 2 LLM wiring rather than faked. Courier API integration, WhatsApp/SMS sending, and a full CRM are architected with clear TODO markers, not pretended complete.

2. Feature Adoption Matrix

Adapting the Group's generic e-commerce blueprint (marketplace-style, Takealot/Sixty60 reference) to Aluminium Store's single-vendor, configurator-led model.

Blueprint Feature Area	Decision	Notes
Multi-vendor marketplace tooling	Drop	Single-vendor platform — no seller onboarding/payouts/commissions needed
Storefront, cart, checkout	Adapt	Configurator-first browsing, not generic PLP-only
Instant EFT / BNPL / card payments	Adopt	8 payment methods modelled; live gateways confirmed per docs/22
Courier + PUDO network	Adopt	Courier/PickupPoint modelled; live API integration is Phase 2
WhatsApp / SMS engagement	Adopt (scaffold)	Provider-adapter pattern built; real provider wiring is Phase 2
CMS + Blog + Help Center	Adopt	Custom CMS, not a third-party tool — matches Roofsteel/Bricksplaza precedent
Full OMS / PIM / CRM	Adapt	Core models built; full automation (auto-dispatch, CLTV) flagged Phase 2
Loyalty & rewards	Adopt	Mechanism real; point/tier values are placeholder assumptions for Fortune to set
Social commerce feeds	Defer	Product.feedExcluded flag ready; feed generation not yet built
Drag-and-drop landing page builder	Defer	ContentBlock CMS covers v1 needs; a builder is its own product

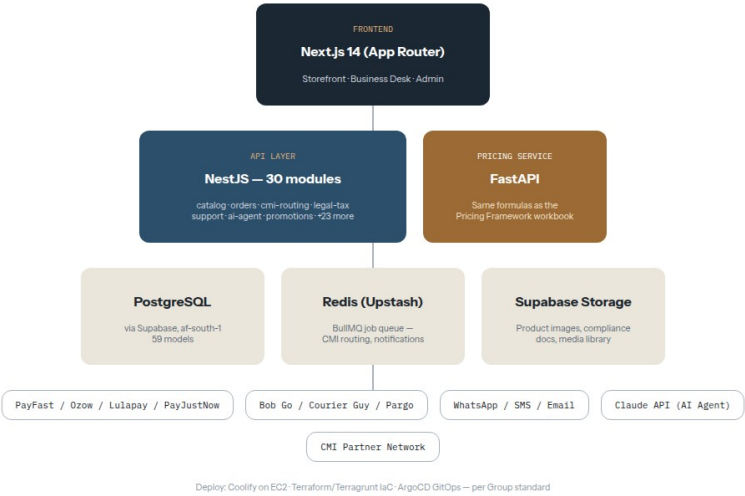
3. Technology Stack

Layer	Choice	Why
Frontend	Next.js 14 (App Router), TypeScript	SSR for catalogue SEO; Group front-end convention
API	NestJS (TypeScript)	Modular; matches Roofsteel/Bricksplaza
Pricing microservice	FastAPI (Python)	Reuses the Pricing Framework's own formula logic directly
Database	PostgreSQL via Supabase, af-south-1	Group standard; data sovereignty
Job queue	BullMQ on Upstash Redis	CMI routing, notifications, price-drop alerts
File storage	Supabase Storage	Product images, compliance certificates
Deploy	Coolify on EC2	Group standard
IaC / GitOps	Terraform/Terragrunt, ArgoCD	Group standard

Estimated scope: ~288 files, ~14,000–18,000 LoC for the Phase 1 scaffold layer — benchmarked against Bellwether SWE Plumbers' actual 35,521 LoC production codebase and adjusted for this platform's larger module count.

4. System Architecture

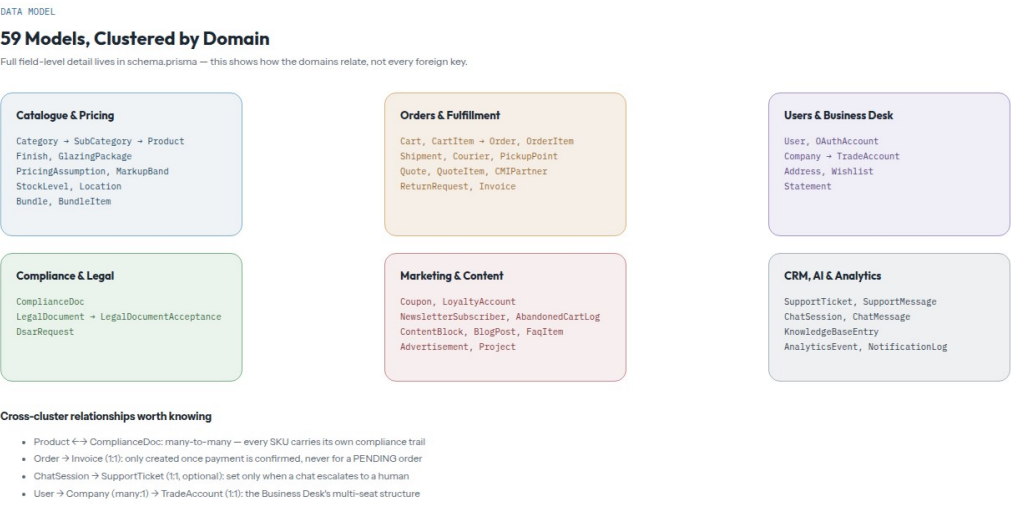
SYSTEM ARCHITECTURE
Aluminium Store Platform



Frontend → API layer → pricing microservice → data layer → external integrations

5. Data Model

59 models across six domains. Full field-level detail lives in the Prisma schema — this diagram shows how the domains relate.

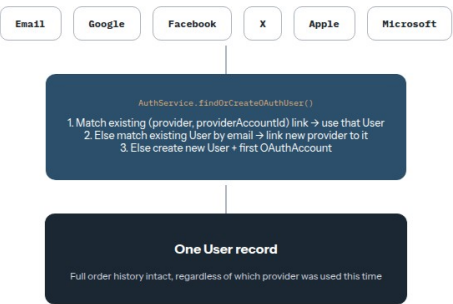


6. Authentication & Single Sign-On

Six sign-in options — Email/password plus Google, Facebook, X, Apple, and Microsoft — all resolving to one User record via account-linking, not six separate identity silos.

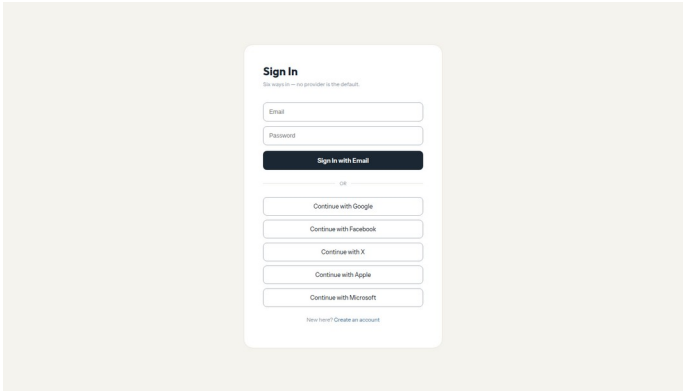
FLOW DIAGRAM

Authentication — 6 Options, 1 Account



Known edge cases, handled explicitly: X does not reliably return a verified email (buyer is prompted to confirm one before checkout). Apple only returns the buyer's name on their very first authorization ever — captured then or not at all.

Account-linking logic — see `AuthService.findOrCreateOAuthUser()`



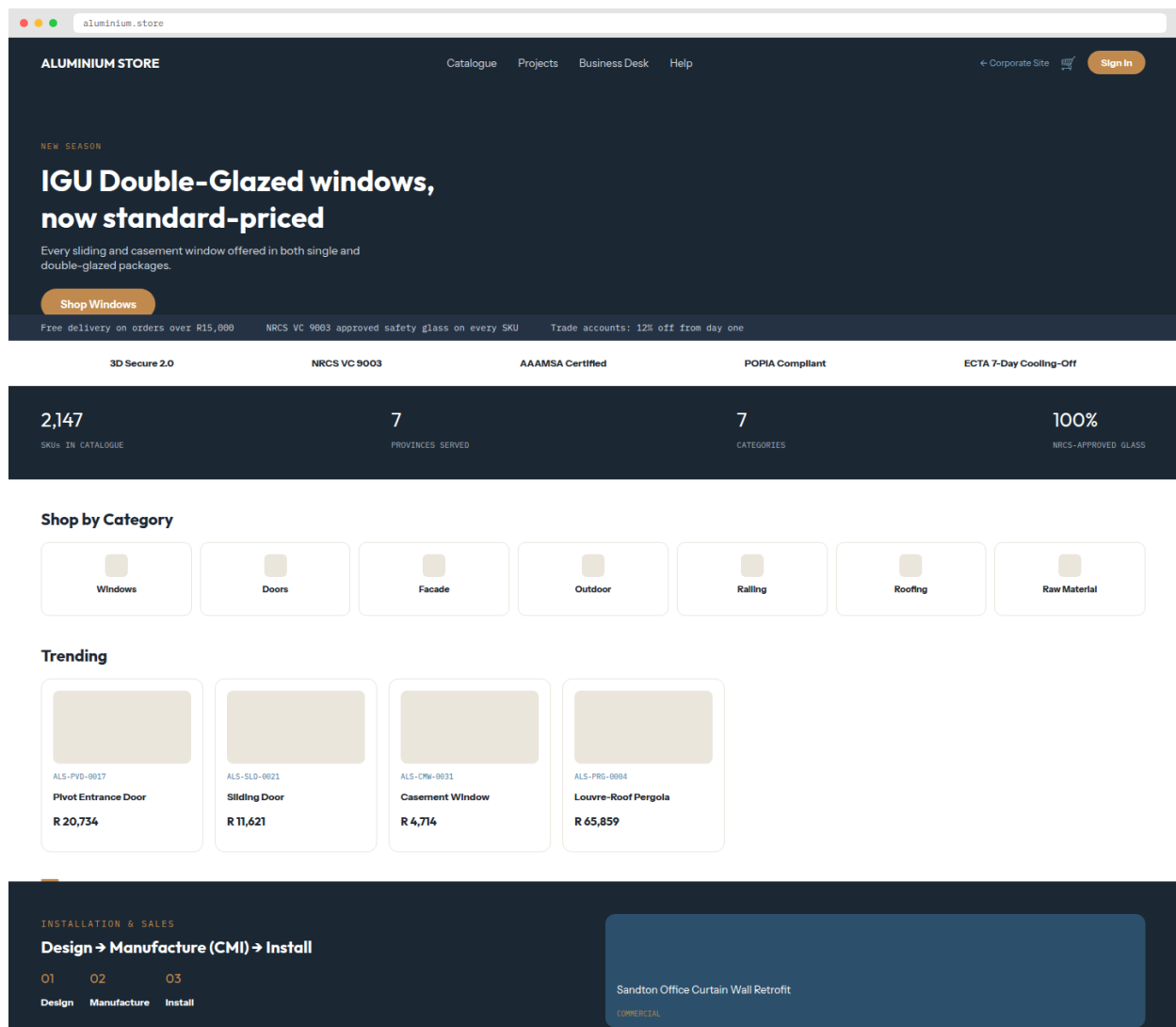
Sign-in screen — all six options rendered as equal-weight buttons

7. Wireframes & Mockups

High-fidelity mockups in the Skyline design system — not grayscale wireframes — so stakeholders see the actual intended look, not an abstraction of it.

7.1 Homepage

18 sections: hero slider (10 rotating promotional slides), announcement ticker, trust badges, animated stat counters, category grid, sector browse, Trending/Recent Arrivals/Clearance carousels, Bundles, 3 ad placements, the Installation/CMI showcase with its projects hero slider, reviews, newsletter signup, and a Business Desk CTA.



Homepage — full section stack

7.2 Product Detail + Configurator

The platform's centrepiece interaction. Size sliders are bounded to the AAAMSA-tested range for that configuration; going outside it routes to an RFQ rather than silently pricing an untested size. Price updates live on every change, calling the same formula verified against the Pricing Framework workbook.

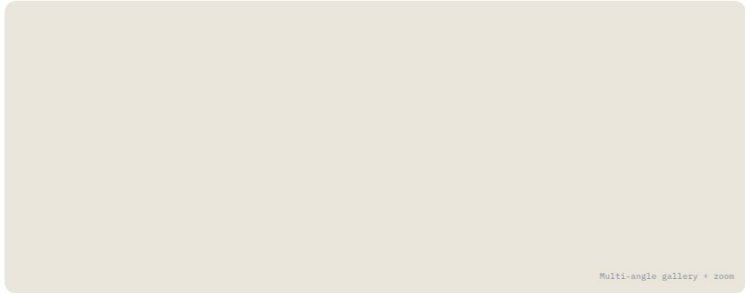
aluminium.store/product/ALS-SLW-0021

ALUMINIUM STORE

ALS-SLW-0021

Aluminium Sliding Window

2-Pane Slider (1 Fixed + 1 Sliding) — AAAMSA-tested up to 3009×1509mm



Multi-angle gallery + zoom



Product Questions

"Does this come with a security lock as standard?" — Answered by Aluminium Store

[Ask a question →](#)

Reviews

★★★★★

"Configurator made pricing painless."

R 3,944.27

Add to Cart

CONFIGURATOR

Width (mm) — max 3009

1209mm

Height (mm) — max 1509

909mm

Finish

Natural Anodised

Glazing Package

Standard Single-Glazed

R 3,944.27

RETAIL

Priced live — matches the Pricing Framework engine exactly

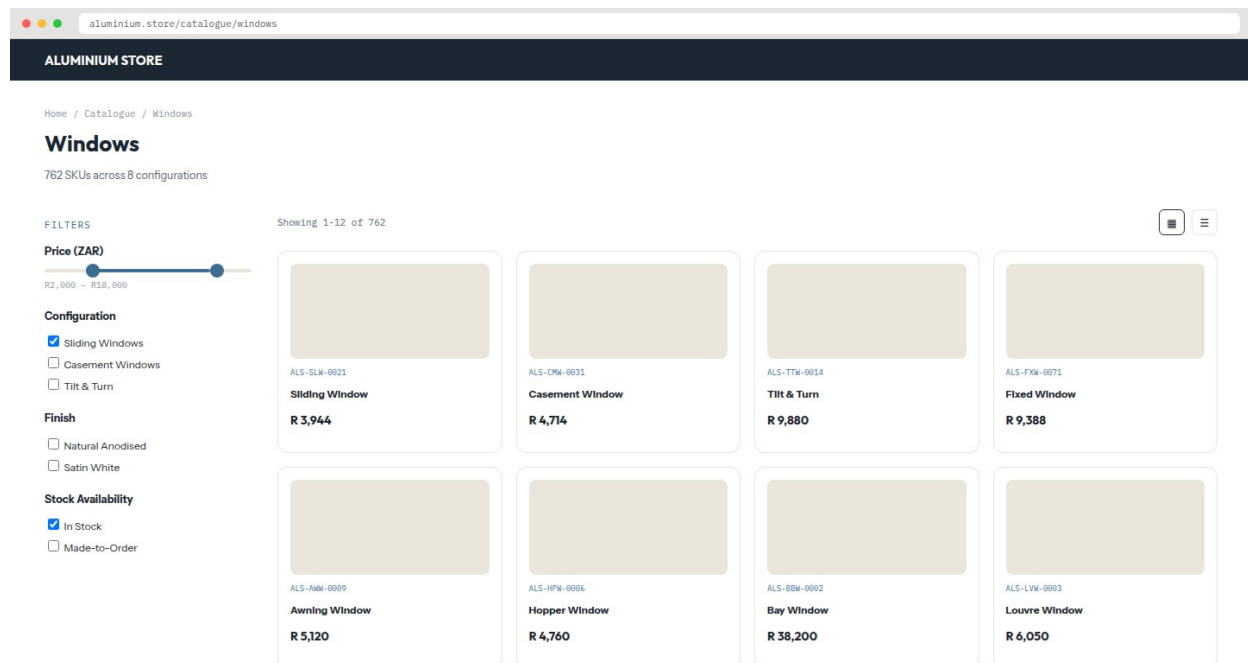
Add to Cart

[Buy Now →](#)

Product + Configurator — live pricing, finish swatches, glazing selection

7.3 Category Browse (PLP)

Faceted filtering — price range, configuration, finish, stock availability — plus grid/list toggle and breadcrumb navigation.



Category page — faceted filters and product grid

7.4 Cart & Checkout

Explicit VAT breakdown (15%, per current SARS rate), coupon field, delivery + fragile-handling surcharge shown as separate line items, pickup-point selector, and a 'Request a Quote Instead' path that converts the cart into a formal RFQ rather than an immediate purchase.

aluminium.store/checkout

ALUMINIUM STORE

1 DELIVERY

2 PAYMENT

3 CONFIRM

Delivery Address

Suburb / postal code autocomplete

ProvinceCity

Pickup Point (optional)
Pargo / PostNet network — map selector

Payment Method

Card / 3DS

Ozow

PayJustNow

SnapScan

☒ I accept the Terms & Conditions and Privacy Policy

ORDER SUMMARY

2x Sliding Window

1x Sliding Door

Coupon code

Apply

Subtotal

VAT (15%)

Delivery

Fragile Handling (8%)

Total

R 7888.54

R 11621.00

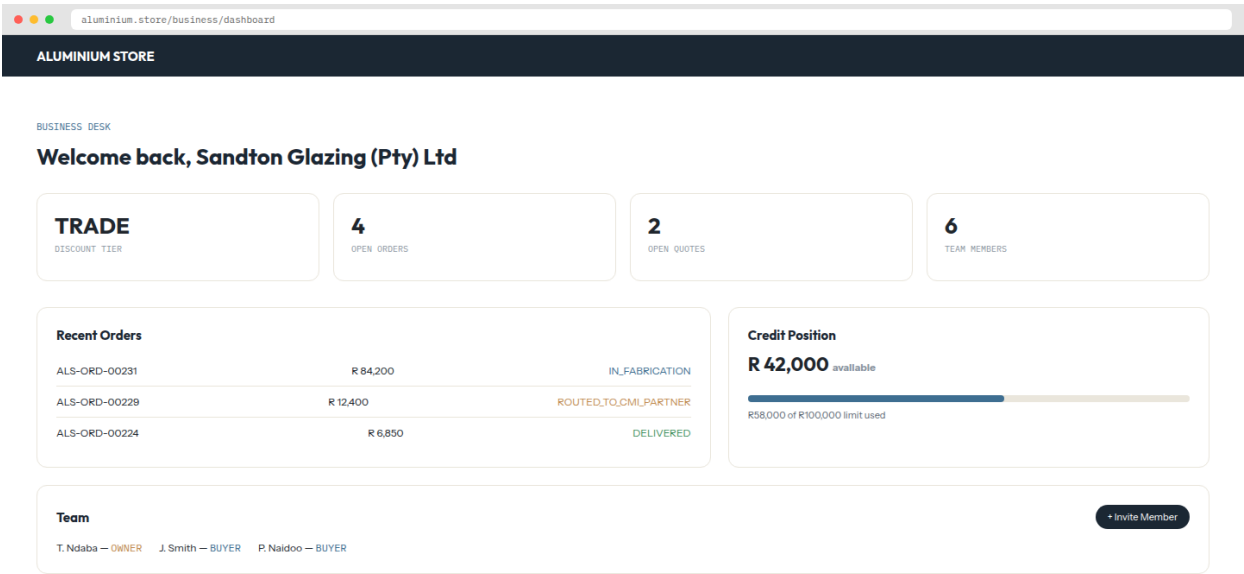
Place Order

Request a Quote Instead →

Checkout — delivery, payment, and order summary

7.5 Business Desk Dashboard

The B2B self-service portal — discount tier, open orders/quotes, credit position, and OWNER-only team management, all scoped to the Company rather than a single User.



Business Desk — company-wide dashboard

7.6 Admin / CMS

Staff-only console: content-block publishing workflow (draft → published), media library, and the trade-account approval queue.

ADMIN

Dashboard

Catalogue Sync

CMS Content

Media Library

Trade Accounts

Quotes

CMI Routing

Compliance Docs

Support Tickets

CMS · CONTENT BLOCKS

Homepage Hero Slides

KEY	TITLE	TYPE	STATUS
hero-slide-1	IGU Double-Glazed windows...	HERO_SLIDE	Published Edit
hero-slide-2	Bundle Deal — door + hardware...	HERO_SLIDE	Published Edit
promo-banner-top	Black Friday — 20% off clearance	PROMO_BANNER	Draft Edit

Pending Trade Account Approvals

Sandton Glazing (Pty) Ltd

Applied 2 days ago

Approve

Review

Admin console — CMS content and trade account approvals

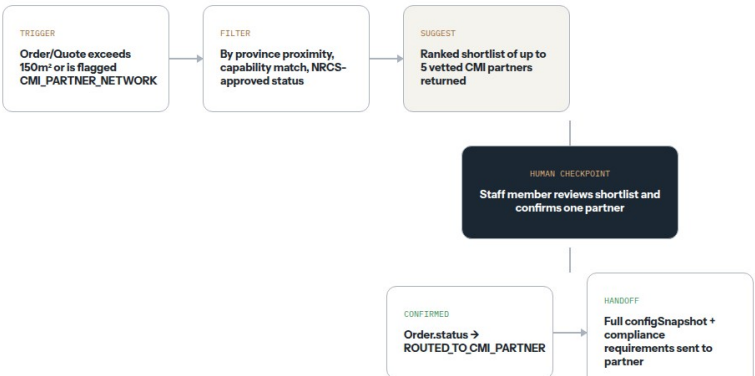
8. CMI Partner Routing

Large curtain-walling and shopfront orders exceed any single hub's fabrication capacity. Routing suggests a ranked shortlist of vetted partners; a staff member always confirms the assignment — the routing engine never auto-assigns, per the Group's 'AI proposes, humans approve' governance principle.

FLOW DIAGRAM

CMI Partner Routing

"AI proposes, humans approve" — the routing engine suggests, a staff member always confirms.



CMI routing flow — trigger → filter → suggest → human checkpoint → handoff

9. Pricing Engine Integration

The FastAPI pricing microservice runs the exact Area Rate formulas verified in the Pricing Framework workbook — not a reimplementation. Tested directly: a 1209×909mm sliding window computes to R3,944.27 retail via both the workbook and the API, to the cent.

- $\text{Base Cost} = \text{opening area} \times (\text{Area Rate for that sub-category} + \text{glazing upgrade}) + \text{hardware allowance}$
- $\text{Retail} = \text{Base Cost} \times (1 + \text{category markup})$; $\text{Trade} = \text{Retail} \times 0.88$; $\text{Volume} = \text{Retail} \times 0.80$
- A scheduled job re-imports the workbook's assumption sheets on change — a pricing-team edit in Excel is the actual change-control mechanism

10. Legal & Tax Compliance

Requirement	How it's enforced
ECTA 7-day cooling-off + Section 43 disclosure	LegalDocument (terms), version tracked per acceptance
ECTA s42(2) non-returnable exclusion	OrderItem.nonReturnable, re-checked server-side by ReturnsService
POPIA privacy notice + Data Subject rights	LegalDocument (privacy), DsarRequest — never auto-approved for deletion
PAIA manual	LegalDocument (paia-manual) — static page + downloadable PDF
CPA 6-month implied warranty	LegalDocument (returns) disclosure text
SARS Tax Invoice	Invoice model — sequential numbering, 15% VAT breakdown, both parties' VAT numbers

VAT confirmed at 15% as of the 25 February 2026 Budget — the 2025-proposed increases to 15.5%/16% were both reversed and not revived. This is a checked fact, not an assumption; re-verify in any future financial year.

Who needs to sign off before launch: a POPIA-accredited Information Officer and the Group's tax advisor. This specification describes what the platform enforces, not a substitute for that review.

11. Backend Module Reference

30 NestJS modules, each with its own guideline doc in the platform's docs/ folder. Grouped by concern:

Domain	Modules
Catalogue & Configurator	catalog · configurator · cms
Cart, Orders & Fulfilment	cart · orders · delivery · logistics · returns
Trade, Quotes & CMI	trade-accounts · quotes · cmi-routing · business-desk
Compliance & Legal	compliance-docs · legal-tax
Payments & Communications	payments · communications
Marketing & Loyalty	promotions · loyalty · newsletter · advertisements
Content	blog · faq · projects · wishlists · product-questions
CRM & AI	support · ai-agent
Platform	auth · admin · analytics

12. Frontend Route Reference

38 routes across the storefront, Business Desk, and admin console.

Area	Key Routes
Storefront	/ · /catalogue · /catalogue/[category]/[subcategory] · /product/[sku] · /cart · /checkout
Content	/about · /blog · /blog/[slug] · /help · /contact · /store-locator · /projects · /cmi-partners
Legal	/legal/[slug] (terms, privacy, cookies, returns, paia-manual, accessibility)
Account	/account · /account/orders · /account/orders/[orderNumber] · /account/trade · /account/addresses
Business Desk	/business/dashboard · /orders · /quotes · /statements · /team · /credit
Auth	/login · /register
Admin	/admin · /catalogue · /trade-accounts · /quotes · /cmi-routing · /compliance-docs
Quotes	/quote/new · /quote/[quoteNumber]

13. Business Desk — Detail

A B2B self-service portal built around a Company model rather than a single User — because a real trade customer is rarely one person. The owner applies, buyers place orders, a bookkeeper needs statements.

Roles

- OWNER — everything below, plus team management (invite/remove) and credit visibility
- BUYER — place orders, submit quotes/RFQs, view own order history
- VIEWER — read-only order history and statements, no ordering

Application flow

POST /trade-accounts/apply creates the Company and an unapproved TradeAccount in one step, setting the applying user as OWNER. Approval remains a manual admin review — not instant — consistent with the same discipline already applied to CMI partner confirmation and DSAR deletion requests: high-consequence actions get a human checkpoint.

14. CMS & Content Management

Two distinct CMS concerns, deliberately not merged into one system:

- Catalogue CMS — manages SKU merchandising fields (images, descriptions, active/inactive), syncing from the Master Product Catalogue workbook rather than allowing free-form SKU invention
- Marketing CMS (ContentBlock) — hero slides, promo banners, category features, SEO metadata, and a shared media library, with an explicit draft → published workflow so a half-finished edit never accidentally goes live

Blog and FAQ sit alongside the CMS as their own simple models (BlogPost, FaqItem) rather than generic content blocks — deliberately not over-engineered with a block-editor for what is, in practice, a title/body/author record.

15. Marketing Engine

Promotions

`PromotionsService.validateCoupon()` is the single source of truth for discount math — cart and checkout call this rather than re-implementing percentage/fixed/free-shipping logic client-side.

Loyalty

Points-per-Rand and tier thresholds are explicitly flagged as placeholder assumptions in the code — real business decisions for Fortune to set, not technical choices baked in silently.

Newsletter & Abandoned Cart

Straightforward subscribe/unsubscribe, plus an abandoned-cart logging hook a scheduled job can use to trigger recovery messaging once the communications providers are live.

16. AI Support Agent — Scope & Architecture

Built to know products, pricing, lead times, technical specifications, and when to hand off to a human — with an important distinction between what's real and what's marked for Phase 2.

Real today

- Full conversation persistence (ChatSession/ChatMessage)
- Keyword-based escalation triggers that create a genuine, linked SupportTicket — a chat that escalates hands a human agent the full transcript, not a cold start

Marked for Phase 2, not faked

- The generative reply itself — currently a structured, honest placeholder rather than a plausible-sounding fabrication
- Vector-similarity knowledge retrieval — currently a naive substring match against KnowledgeBaseEntry, proving the data flow without pretending to be production-grade search

Recommended real implementation: Claude (Anthropic) via the API, grounded by pgvector similarity search over KnowledgeBaseEntry rows seeded from the FAQ, Pricing Framework, and Compliance & Regulatory Framework — each entry carrying a sourceRef for traceability.

17. Logistics & Communications

Logistics

Two distinct concerns: delivery (fee/zone calculation, weight-and-distance banded per the Pricing Framework) and logistics (the courier directory and PUDO pickup-point network — Pargo/PostNet). Fragile-glazed shipments default to Besfleet, the Group's own fleet, rather than a general parcel courier.

Communications

A provider-adapter pattern mirroring the payments module — one method per channel (WhatsApp, SMS, Email), each logging to NotificationLog with a QUEUED status rather than silently doing nothing while the real provider integration is pending.

18. Service Provider Recommendations

Need	Recommended
Additional Instant EFT	Stitch, Capitec Pay
Additional BNPL	PayFlex, Mobicred
WhatsApp Business API	Twilio, Bird, or Infobip
SMS Gateway	Clickatell or BulkSMS (both SA-based)
Courier Aggregator	Bob Go (formerly uAfrica), ShipLogic
Search (if Postgres full-text proves insufficient)	Meilisearch (self-hostable) or Algolia
Behaviour Analytics	PostHog (self-hosted) or GA4
AI Agent LLM + Vector Store	Claude (Anthropic) + pgvector
Helpdesk (if scale justifies buying vs. building further)	Freshdesk or Zendesk
POPIA / PAIA / SARS review	A POPIA-accredited Information Officer and the Group's tax advisor — not a software decision

19. Roadmap — HITLAD Phases

- Phase 1 — Foundation: schema, auth, catalog read APIs, basic storefront browsing (this scaffold)
- Phase 2 — Configurator & Pricing: FastAPI pricing microservice live, configurator UI, cart
- Phase 3 — Orders, Payments & Trade Accounts: checkout, live payment gateways, trade account approval flow
- Phase 4 — CMI Routing & Compliance: live partner routing, compliance-document attachment, quote/RFQ flow
- Phase 5 — Delivery, Admin & Launch Hardening: courier API integration, admin CMS, security review

20. Open Questions

- CMI hub-capacity threshold — is 150m² per order the right default trigger?
- Loyalty points-per-Rand and tier thresholds — currently placeholder assumptions
- Besfleet booking integration — API/booking system, or manual handoff for now?
- Which courier aggregator (Bob Go vs. direct integrations) for Phase 5?
- POPIA Information Officer and SARS tax advisor sign-off — scheduling this before Phase 3
- Social media handles — LinkedIn, YouTube, and Behance remain registered as placeholders; confirm timeline for making them live